

Chapter 2: Complex Numbers - Formula Sheet

1. Basic Definitions

Concept	Formula	Notes
Complex Number	$z = a + ib$	where $a, b \in \mathbb{R}$ and $i^2 = -1$
Real Part	$\text{Re}(z) = a$	
Imaginary Part	$\text{Im}(z) = b$	
Equality	$z_1 = z_2 \Leftrightarrow a_1 = a_2 \text{ and } b_1 = b_2$	

2. Complex Conjugate

Property	Formula
Definition	$\bar{z} = a - ib$
Sum with conjugate	$z + \bar{z} = 2\text{Re}(z)$
Difference with conjugate	$z - \bar{z} = 2i\text{Im}(z)$
Product with conjugate	$z\bar{z} = (\text{Re}(z))^2 + (\text{Im}(z))^2 = a^2 + b^2$
Conjugate of sum	$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
Conjugate of difference	$\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
Conjugate of product	$\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$
Conjugate of quotient	$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$
Conjugate of power	$\overline{z^n} = (\bar{z})^n$

3. Modulus (Magnitude)

Property	Formula
Definition	$\ z\ = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$
Modulus of conjugate	$\ \bar{z}\ = \ z\ $
Modulus of negative	$\ -z\ = \ z\ $
Inequality with real part	$\ z\ \geq \text{Re}(z)$
Inequality with imaginary part	$\ z\ \geq \text{Im}(z)$
Modulus of product	$\ z_1 z_2\ = \ z_1\ \cdot \ z_2\ $
Modulus of quotient	$\left\ \frac{z_1}{z_2}\right\ = \frac{\ z_1\ }{\ z_2\ }$
Modulus of power	$\ z^n\ = \ z\ ^n$
Triangle Inequality	$\ z_1 + z_2\ \leq \ z_1\ + \ z_2\ $

4. Operations on Complex Numbers

Operation	Formula	Result
Addition	$(a_1 + ib_1) + (a_2 + ib_2)$	$= (a_1 + a_2) + i(b_1 + b_2)$

Operation	Formula	Result
Subtraction	$(a_1 + ib_1) - (a_2 + ib_2)$	$= (a_1 - a_2) + i(b_1 - b_2)$
Multiplication	$(a_1 + ib_1)(a_2 + ib_2)$	$= (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)$
Division	$\frac{z_1}{z_2} = \frac{z_1 \overline{z_2}}{\ z_2\ ^2}$	$= \frac{a_1a_2 + b_1b_2 + i(a_2b_1 - a_1b_2)}{a_2^2 + b_2^2}$
Reciprocal	$\frac{1}{z} = \frac{1}{a + ib}$	$= \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$

5. Forms of Complex Numbers

5.1 Algebraic Form

$$z = a + ib$$

5.2 Trigonometric (Polar) Form

Component	Formula	Meaning
Modulus	$r = \ z\ = \sqrt{a^2 + b^2}$	Distance from origin
Argument	$\theta = \arg(z) = \tan^{-1}\left(\frac{b}{a}\right)$	Angle with positive real axis, $-\pi < \theta \leq \pi$
Real part in polar	$a = r\cos(\theta)$	
Imaginary part in polar	$b = r\sin(\theta)$	
Polar form	$z = r(\cos(\theta) + i\sin(\theta))$	

5.3 Exponential Form (Euler's Formula)

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$z = re^{i\theta}$$

Euler's Identity:

$$e^{i\pi} + 1 = 0$$

6. Operations in Polar Form

Operation	Input	Output
Multiplication	$z_1 = r_1(\cos\theta + i\sin\theta)$ $z_2 = r_2(\cos\phi + i\sin\phi)$	$z_1z_2 = r_1r_2[\cos(\theta + \phi) + i\sin(\theta + \phi)]$
Division	$z_1 = r_1(\cos\theta + i\sin\theta)$ $z_2 = r_2(\cos\phi + i\sin\phi)$	$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta - \phi) + i\sin(\theta - \phi)]$

7. De Moivre's Formula

$$z^n = r^n [\cos(n\theta) + i\sin(n\theta)]$$

or in exponential form:

$$z^n = r^n e^{in\theta}$$

8. Euler's Formula Consequences

Expression	Formula
Conjugate in exponential	$e^{-i\theta} = \cos(\theta) - i\sin(\theta) = \overline{e^{i\theta}}$
Cosine formula	$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$
Sine formula	$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

9. Square Roots of Complex Numbers

For $z = a + ib$, let $w = x + iy$ where $w^2 = z$

System of Equations:

$$x^2 - y^2 = a$$

$$2xy = b$$

$$x^2 + y^2 = \sqrt{a^2 + b^2}$$

Solutions:

Condition	Root 1 (w_1)	Root 2 (w_2)
If $b > 0$	$w_1 = \sqrt{\frac{a + \ z\ }{2}} + i \sqrt{\frac{\ z\ - a}{2}}$	$w_2 = -w_1$
If $b < 0$	$w_1 = \sqrt{\frac{a + \ z\ }{2}} - i \sqrt{\frac{\ z\ - a}{2}}$	$w_2 = -w_1$
If $b = 0$	$w_1 = \sqrt{a}$	$w_2 = -\sqrt{a}$

10. Quadratic Equation Solutions

For $az^2 + bz + c = 0$ where $a, b, c \in \mathbb{C}$

Discriminant: $\Delta = b^2 - 4ac$

Case	Condition	Solutions
Case 1	$\Delta \in \mathbb{R}, \Delta > 0$	$z_1 = \frac{-b + \sqrt{\Delta}}{2a}, \quad z_2 = \frac{-b - \sqrt{\Delta}}{2a}$
Case 2	$\Delta \in \mathbb{R}, \Delta < 0$ Let $\Delta = -k$ where $k > 0$	$z_1 = \frac{-b + i\sqrt{k}}{2a}, \quad z_2 = \frac{-b - i\sqrt{k}}{2a}$
Case 3	$\Delta \notin \mathbb{R}$ Find square roots r_1, r_2 of Δ	$z_1 = \frac{-b + r_1}{2a}, \quad z_2 = \frac{-b + r_2}{2a}$

11. The n-th Roots of Complex Numbers

For $z = r(\cos\theta + i\sin\theta)$, the n roots of z are:

$$w_k = r^{1/n} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right) \right]$$

where $k = 0, 1, 2, \dots, n-1$

In exponential form:

$$w_k = r^{1/n} e^{i(\theta + 2k\pi)/n}$$

12. Fundamental Theorem of Algebra

Every polynomial of degree n :

$$P(z) = c_0 + c_1z + c_2z^2 + \dots + c_nz^n$$

has exactly n complex roots (counting multiplicities), and can be factored as:

$$P(z) = (z - z_0)(z - z_1)(z - z_2) \dots (z - z_{n-1})$$

13. Important Identities

Identity	Formula
Sum/Difference formulas	$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$ $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$
Product identity	$e^{i\theta} \cdot e^{i\phi} = e^{i(\theta+\phi)}$
Useful relation	$\overline{wz} + w\overline{z} = 2\operatorname{Re}(wz)$

Quick Reference: Key Properties

$$\sqrt{\overline{z}} = z \quad \sqrt{z} = -z \implies \operatorname{Re}(z) = 0 \text{ (purely imaginary)}$$

$$\sqrt{|1 + iz|} = \sqrt{|1 - iz|} \implies z \in \mathbb{R}$$

$$\sqrt{z\overline{z}} = z + 3\overline{z} \implies z \text{ is real}$$